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SEAT No. :

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OCT. -16/BE/Insem. - 123

B.E. (Mechanical Engg.)

OPERATION RESEARCH

(2012 Course) (Semester - I) (Elective-II) (402045C)

Time : 1 Hour]

[Max. Marks :30

Instructions to the candidates:

- 1) Neat diagrams must be drawn wherever necessary.*
- 2) Figures to the right side indicate full marks.*
- 3) Use of logarithmic tables slide rule, Mollier charts, electronic pocket calculator and steam tables is allowed.*
- 4) Assume suitable data, if necessary.*

Q1) Solve the following LPP by simplex method

[10]

Maximize $Z = 3x_1 + 2x_2$

subject to: $x_1 + x_2 \leq 4$

$x_1 - x_2 \leq 2$

$x_1, x_2 \geq 0$

OR

Q2) a) Define the following:

[4]

- i) Decision tree.
 - ii) Decision node.
 - iii) Laplace criterion.
- b) A steel manufacturing company is concerned with the possibility of a strike. It will cost an extra Rs. 20,000 to acquire an adequate stockpile. If there is a strike and the company has not stockpiled, management estimates an additional expenses of Rs. 60,000 on account of lost sales. Should the company stockpile or not if it is to use Laplace Criterion?

[6]

P.T.O.

Q3) Obtain Initial basic feasible solution (IBFS) for following transportation problem and use MODI method to test optimality of transportation problem. [10]

	D1	D2	D3	D4	Supply
Plant	2	3	11	7	6
	1	0	6	1	1
	5	8	15	9	10
Demand	7	5	3	2	

Use VAM to find IBFS

OR

Q4) a) Five jobs are to be assigned to five machines to minimize the total time required to process the jobs on machines. The times in hours for processing each job on each machine are given in the matrix below. By using assignment algorithm make the assignment for minimizing total time of processing. [6]

	m/c A	m/c B	m/c C	m/c D	m/c E
Job 1	2	4	3	5	4
Job 2	7	4	6	8	4
Job 3	2	9	8	10	4
Job 4	8	6	12	7	4
Job 5	2	8	5	8	8

b) Differentiate between transportation problem and assignment. [4]

Q5) Find the solution for game where pay-off matrix is given in table below. [10]

Person A	Person B	
	7	4
	5	6

OR

Q6) a) Define the following:

[4]

- i) Saddle point.
- ii) Pay-off matrix.
- iii) Dominance principle.
- iv) Value of game.

b) A manufacturing firm produces a single product whose selling price is Rs. 16 per unit and the variable cost per unit is Rs. 12. If the annual fixed cost of the firm are estimated as Rs. 1,20,000. Find the break even points in units, in rupees and as a percentage of capacity in the firm has an estimated capacity of 50,000 units of the product. What is the margin of safety? **[6]**

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